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Artificial Intelligence

Improving Generalization of Deep Learning for Glaucoma Classification Under Real-World Domain Shift via Unsupervised Domain Adaptation

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Purpose: To develop and evaluate an unsupervised domain adaptation (UDA) framework for glaucoma classification from fundus images that improves the generalizability of deep learning (DL) models across heterogeneous imaging characteristics and clinical settings.

Methods: We developed an adversarial UDA framework that adapts a labeled source domain to an unlabeled target domain by jointly optimizing glaucoma classification and domain discrimination. A total of 6906 fundus images were included: 1422 images (full-view and optic nerve head-cropped) derived from 711 fundus photographs of 520 patients from the University of Illinois Chicago and 5484 images from three public datasets (RIMONE-DL, $n = 371$; REFUGE, $n = 259$; LAG, $n = 4854$). Performance was evaluated across multiple source–target domain pairs. Saliency map analysis compared feature utilization between UDA and standard DL models.

Results: Across diverse source–target dataset pairs, the proposed UDA framework improved standard DL classification accuracy by up to 40.0% and area under the curve (AUC) by up to 59.8% in extreme domain shift experiments. UDA also reduced the performance gap to the ideal target-trained model by up to 82.6% for accuracy and 38.1% for AUC. When applied to unseen domains, UDA improved classification accuracy by up to 14.7% relative to standard DL.

Conclusions: The proposed UDA framework improves cross-domain generalizability of DL-based glaucoma classification from fundus images. Compared with standard DL models, UDA exhibits feature utilization patterns more aligned with an ideal baseline, reducing overreliance on optic nerve head–specific cues and supporting more robust decision-making under domain shift.

Translational Relevance: By enabling glaucoma classification without requiring labeled data from new clinical sites, this UDA framework addresses a key barrier to deploying artificial intelligence–based screening tools across diverse real-world ophthalmic settings.

Introduction

Glaucoma is a prevalent sight-threatening eye condition that is difficult to detect early due to its asymptomatic nature, leading to a 50% undetected rate until significant vision loss occurs.[1–4] Timely diagnosis is crucial to prevent irreversible blindness.[1,5,6] Recent advancements in deep learning (DL) for computer vision have shown promise in glaucoma screening and diagnosis using fundus images.[7–14] DL models have demonstrated strong performance in predicting glaucoma through direct classification, segmentation of key fundus regions (e.g., optic cup, optic disc,

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blood vessels), or multitask learning combining these tasks.[7–14] However, a key limitation hindering clinical translation is their reduced ability to generalize across diverse datasets, particularly when applied to data that differ from the training distribution, a problem commonly referred to as domain shift.[14–18]

Domain shift is a pervasive challenge in real-world medical imaging, arising from variations in imaging protocols, acquisition devices, and patient demographics.[19] For example, fundus images collected at two different hospitals may represent distinct domains, often referred to as a source domain (used for training) and a target domain (used for deployment), due to differences in camera types, imaging workflows, and population characteristics. DL models trained on one domain often struggle to maintain performance when applied to another, resulting in significant performance degradation.[20] This lack of generalization remains a key barrier for clinical deployment and poses substantial risks, including missed or incorrect diagnoses.

Increasing attention has been directed toward addressing domain shift in artificial intelligence–based glaucoma diagnosis, with several studies proposing strategies to enhance model generalization across diverse imaging conditions. For instance, Chuter et al.[20] validated the RETFound foundation model on a large, heterogeneous fundus imaging dataset and demonstrated robust performance in binary glaucoma classification across varied demographic subgroups. Similarly, Fan et al.[21] introduced a vision transformer–based approach to improve the generalizability of state-of-the-art convolutional DL models in cross-domain glaucoma classification. While these architectural and pretraining advances represent important steps forward, they may still fall short in scenarios with pronounced domain discrepancies, particularly when imaging conditions, population characteristics, or devices differ substantially between training and deployment settings.

To explicitly tackle these domain discrepancies, domain adaptation techniques have emerged as a complementary solution for improving model robustness.[22] Given that label collection in medical imaging is labor-intensive and requires expert grading, unsupervised domain adaptation (UDA), which does not rely on labeled data from the target domain, has gained increasing attention in glaucoma classification.[23] UDA methods aim to reduce distributional mismatches by leveraging unlabeled target domain images during training.[24] Recent studies have demonstrated that UDA can effectively mitigate domain shifts in fundus-based glaucoma diagnosis, improving model performance across diverse clinical datasets.[25] Two principal categories of UDA techniques in this context are (1) feature-level alignment, often using adversarial learning to harmonize latent feature distributions, and (2) image-level alignment, typically using generative adversarial networks (GANs),[25,26] to reduce visual variability by transforming images across domains.[24]

Liu et al.[27] and Zhou et al.[28] proposed adversarial UDA frameworks for joint glaucoma classification and optic cup/disc segmentation, demonstrating improved generalization across datasets. However, these methods require segmentation annotations in the source domain, which is a labor-intensive process that requires expert graders, thereby limiting their scalability in practice. In contrast, Sun et al.[29] employed an image-level UDA strategy using GANs to synthesize target-like fundus images, improving classification performance by narrowing domain gaps. While promising, GAN-based methods introduce significant computational overhead and are prone to training instabilities such as mode collapse.[30] Other related studies have applied UDA to broader tasks in glaucoma diagnosis, including segmentation of the optic disc, cup, and vasculature in fundus images, as well as retinal fluid and layer segmentation in optical coherence tomography.[26,31–36]

In this work, we propose a UDA approach for glaucoma classification that operates without requiring labeled target domain data. Building on the adversarial learning framework introduced by Ganin and Lempitsky,[37] our approach encourages the learning of glaucoma-discriminative features while simultaneously minimizing domain-specific information. By jointly optimizing for glaucoma classification and domain discrimination, our model reduces domain discrepancies and improves generalization across varied imaging conditions. We benchmark our method against state-of-the-art standard DL classifiers and evaluate its performance across five fundus image datasets, including three publicly available datasets (REFUGE, LAG, and RIMONE-DL) and two private datasets from the University of Illinois Chicago (UIC).

The key contributions of this study are threefold. (1) We propose a novel adversarial UDA method for glaucoma classification and validate it on diverse fundus imaging datasets. (2) We address the limited generalizability of standard DL models under domain shift by enhancing robustness without requiring target domain labels. (3) We benchmark our method against both standard and idealized DL baselines, demonstrate its ability to significantly narrow the performance gap, and analyze decision-making behavior using Grad-CAM^{++} to assess interpretability and alignment with glaucoma-relevant features. The study source code will be made publicly available on GitHub upon accep-

tance of the manuscript at https://github.com/homairs/Unsupervised_DA.git.

Methods

Study Datasets

We used three public fundus datasets with binary glaucoma labels—LAG,[10] REFUGE,[38] and RIMONE-DL[39]—and a subset of private UIC fundus images, including both full-view and optic nerve head (ONH)–cropped views, collected between 2006 and 2018. We used billing information (International Classification of Diseases [ICD] codes) as binary glaucoma labels for the UIC datasets. A detailed list of ICD codes included for the glaucoma class in the UIC dataset is provided in Supplementary Table S6. We additionally utilized previously collected expert manual glaucoma gradings for a subset of the UIC test images to further validate model performance on this cohort. Expert grading was performed on fundus photographs alone, as previously described by Rashidisabet et al.[40] The UIC data were anonymized, and all methods were carried out in accordance with relevant guidelines and regulations. The study was approved by the University of Illinois Chicago Institutional Review Board, and informed consent was obtained from all participants.[41] The study adhered to the tenets of the Declaration of Helsinki. RIMONE-DL fundus images are naturally cropped around the ONH region. Therefore, we created a cropped version of the UIC dataset around the ONH using the method proposed in Rashidisabet et al.,[42] and we called that dataset UIC_{Crop}. A summary of all study data specifications, including age distribution, number of images per class, glaucoma grading criteria, and data sources for all five datasets (UIC, UIC_{Crop}, REFUGE, LAG, and RIMONE-DL), is provided in Supplementary Table S7. Table 1 summarizes dataset sizes, acquisition devices, and example images to illustrate domain shifts. Each dataset was split into 70% training, 10% validation, and 20% testing, as shown in Table 2. Further, we split the UIC data by patient to prevent information leakage from the test set into the training set. In this study, a source dataset refers to a labeled dataset used for model training, whereas a target dataset refers to a dataset from a different domain, such as a different institution, imaging source, or preprocessing condition, used for adaptation and/or testing in subsequent experiments.

Standard Deep Learning for Glaucoma Classification

We utilized state-of-the-art ResNet18 and ResNet50 architectures[43] as standard DL baselines in glaucoma

Table 1. Data Specification for Fundus Datasets Used in This Study

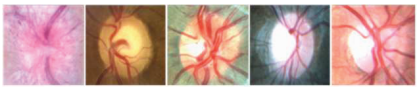
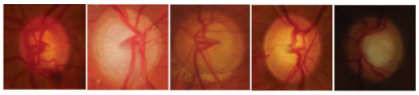
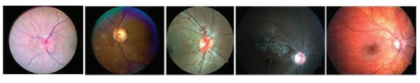
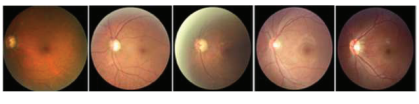
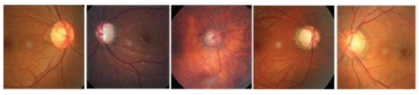
View	Dataset	Size	Acquisition Device	Randomly Sampled Images
ONH crops	UIC_{Crop}	712	1) Zeiss CIRRUS photo 800. 2) Zeiss FF450 fundus camera	
	RIMONE-DL	371	1) Nidek AFC-210 non-mydriatic fundus camera. 2) A non- mydriatic Kowa WX 3D stereo fundus camera	
Full fundus	UIC	712	1) Zeiss CIRRUS photo 800. 2) Zeiss FF450 fundus camera	
	REFUGE	259	1) Zeiss Visucam 500 fundus camera. 2) Canon CR-2 device	
	LAG	4,854	Details unavailable	

Table 2. Train, Validation, and Test Data Split Breakdown for Fundus Datasets Used in this Study

		Number of Total (T), Glaucoma (G), and Nonglaucoma (NG) Images		
View	Data	Train (70%)	Validation (10%)	Test (20%)
ONH crops	UIC_{Crop}	T = 520 G = 250, NG = 270	T = 62 G = 28, NG = 34	T = 129 G = 77, NG = 52
	RIMONE-DL	T = 260 G = 99, NG = 161	T = 37 G = 36, NG = 1	T = 74 G = 37, NG = 37
Full fundus	UIC	T = 520 G = 250, NG = 270	T = 62 G = 28, NG = 34	T = 129 G = 77, NG = 52
	REFUGE	T = 174 G = 71, NG = 103	T = 25 G = 24, NG = 1	T = 50 G = 25, NG = 25
	LAG	T = 3,398 G = 826, NG = 2572	T = 485 G = 485, NG = 0	T = 971 G = 400, NG = 571

classification. To establish competitive baselines, we applied transfer learning by initializing the models with weights pretrained on the ImageNet dataset. This warm start enhances optimization and facilitates effective feature extraction from fundus images for glaucoma classification. All baseline models underwent further training to adapt to the domain-specific features of fundus images and were optimized through comprehensive hyperparameter tuning. A detailed description of the implementation process can be found in Section 2.7.

Domain Adaptation for Glaucoma Classification

Figure 1 illustrates our proposed adversarial UDA method, building on the framework proposed by Ganin and Lempitsky.[37] Consistent with the baseline models, we employed ResNet18 or ResNet50 architectures as the feature extractor backbone, pretrained on the ImageNet dataset. All UDA models were fine-tuned on fundus images and optimized through comprehensive hyperparameter tuning. A detailed description of the implementation process can be found in Section 2.7.

Unlike the baselines, which optimize solely for glaucoma classification, our proposed UDA approach incorporates a shared feature extractor and two output heads: one for glaucoma classification and another for domain discrimination, as shown in Figure 1. This framework is adversarial but not generative; it does not generate synthetic fundus images. Instead, it uses adversarial supervision to learn feature representations that are predictive of glaucoma while being less informative about whether an image originated from the source or target domain. The UDA model uses labeled source data and unlabeled target data during training. The glaucoma classification head is trained using source domain glaucoma labels, whereas the domain discriminator is trained using source-versus-target domain labels. A gradient reversal layer (GRL), proposed by Ganin and Lempitsky,[37] is placed before the domain discriminator. During backpropagation, the GRL reverses the domain discrimination gradients passed to the feature extractor, encouraging the feature extractor to learn domain-invariant representations while preserving glaucoma-discriminative information. Our proposed UDA framework is architecture-agnostic, and the feature extractor backbone can be flexibly replaced with other convolutional or transformer-based networks without modification to the overall training scheme.

Let F denote the feature extractor, C the glaucoma classification head, and D the domain discriminator. For labeled source images (x_s, y_s) and unlabeled target images x_t, the optimization objective can be written as

$$\min_{F,\,C}\max_{D}\mathcal{L}_{cls}\left(C\left(F\left(x_s\right)\right),\ y_s\right)-\lambda_p\mathcal{L}_{dom}\left(D\left(F\left(x\right)\right),d\right)\quad(1)$$

where x includes both source and target images, d denotes the domain label, $\mathcal{L}_{cls}$ is the glaucoma classification cross-entropy loss, $\mathcal{L}_{dom}$ is the domain discrimination cross-entropy loss, and λ_p controls the strength of adversarial adaptation during training. In practice, the GRL implements this minimax objective by acting as an identity function during the forward pass and multiplying the gradient from the domain discriminator by $-\lambda_p$ during backpropagation.

Experimental Setup

As shown in Figure 2, we compared three methods for glaucoma classification: standard DL models, the UDA method, and an ideal model. The standard DL models were trained exclusively on source data without

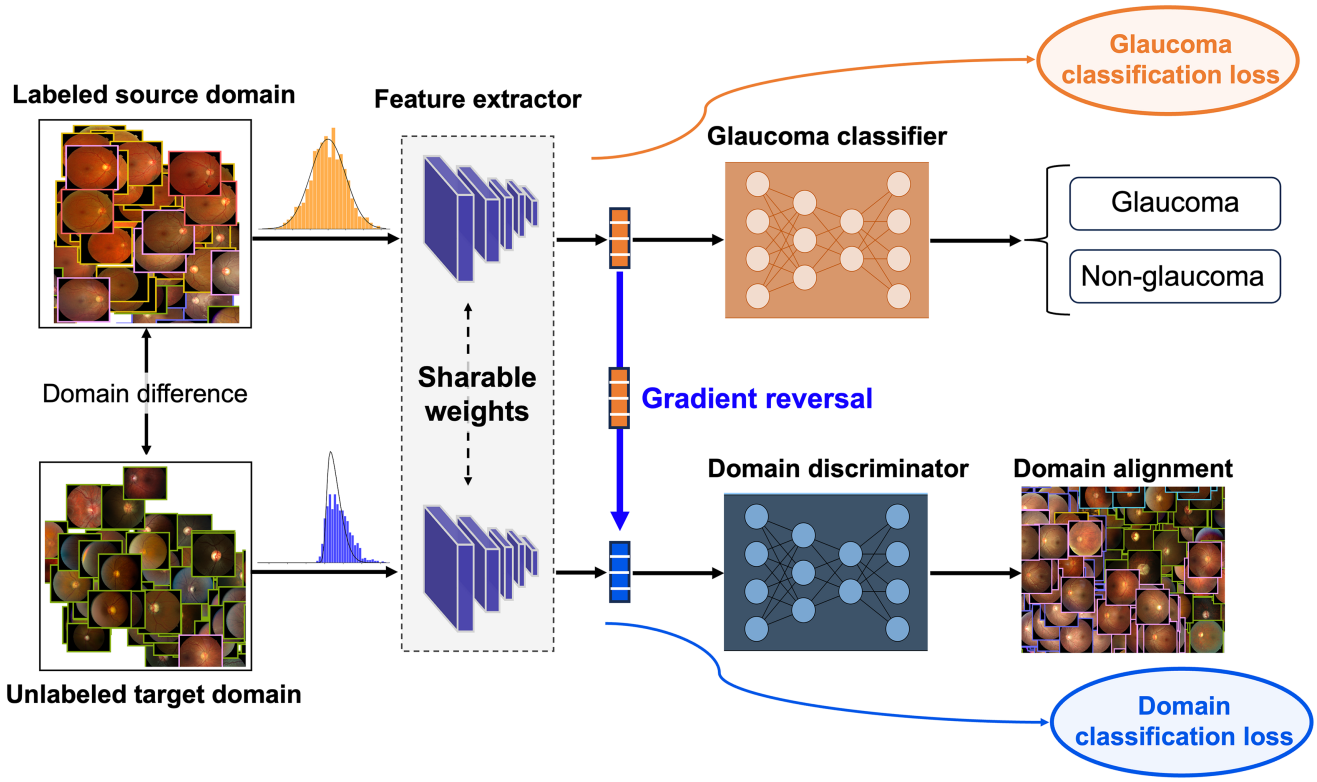


Figure 1. Proposed unsupervised domain adaptation pipeline. The framework simultaneously optimizes binary glaucoma classification and source-versus-target domain discrimination using labeled source and unlabeled target data. A gradient reversal layer adversarially trains the shared feature extractor to preserve glaucoma-discriminative information while reducing domain-specific information, promoting domain alignment and domain-invariant feature learning.

incorporating target data, while the ideal models were trained solely on the target domain with class labels provided. In contrast, our UDA models were trained on labeled source data and an unlabeled target training set, then tested on target test data. To thoroughly evaluate model performance across all possible source–target pairs, we systematically swapped source and target data between all five datasets: UIC, LAG, REFUGE, RIMONE-DL, and UIC_{Crop}.

We developed 10 baseline models, consisting of five standard DL models and five ideal models, using each dataset as either the source or the target. Additionally, we created 20 UDA models by exploring all possible combinations of the five datasets: UIC, UIC_{Crop}, LAG, REFUGE, and RIMONE-DL as both source and target domains. Because the UIC and UIC_{Crop} dataset labels were derived from ICD codes and may contain noise, we further validated the standard DL, UDA, and ideal (train-on-target) models using glaucoma expert manual gradings, as discussed in Section 2.1. Furthermore, we created source + target baseline models, which were trained and validated on the combined data from both the source and target domains. This allowed us to compare the generalizability of the standard DL models versus our UDA method on unseen target domains.

Statistical Analysis

All statistical analyses were performed in Python. Glaucoma classification performance was evaluated using accuracy, sensitivity, precision, F1-score, and area under the receiver operating characteristic curve (AUC), as defined in Supplementary Section 1.4. Performance metrics were reported separately for each source–target dataset configuration. For AUC, pairwise model comparisons within each source–target configuration were performed using DeLong's test, which accounts for the paired nature of predictions generated by different models on the same target test images. To account for within-dataset variability, image-level paired bootstrap analyses were also performed within each source–target configuration for accuracy and AUC differences between models. Bootstrap-based 95% confidence intervals (CIs) were estimated by resampling target test images and calculating the paired performance difference between models. To assess whether performance differences

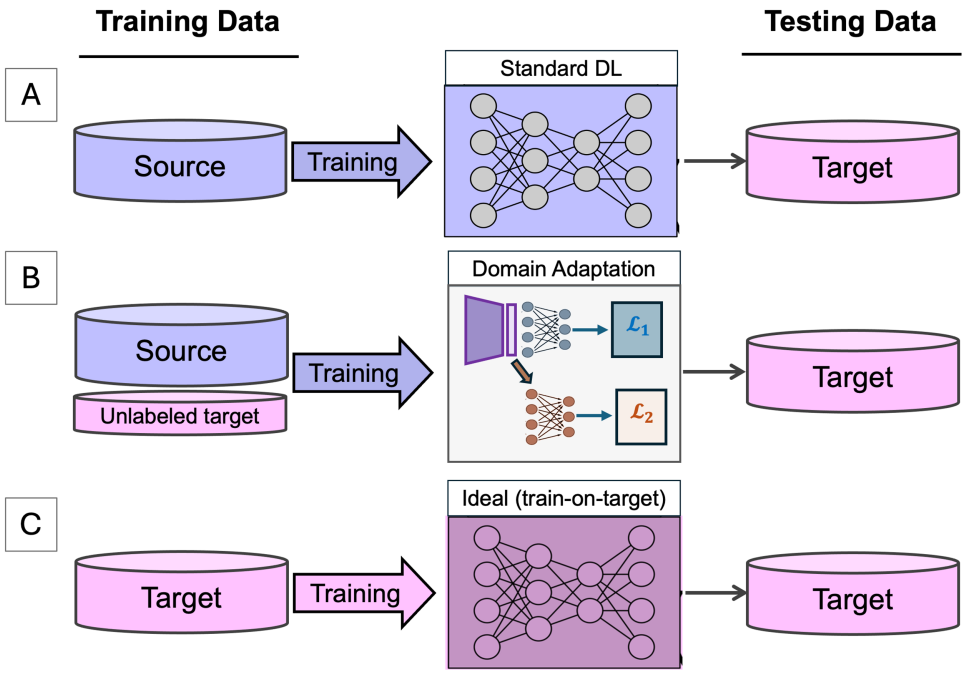


Figure 2. Schematic illustration of experimental setup. (**A**) Standard DL method trained exclusively on source data. (**B**) UDA method trained on source and unlabeled target data. (**C**) Ideal model trained exclusively on target data.

were consistent across source–target configurations, nonparametric paired analyses were performed using the Wilcoxon signed-rank test applied to matched configuration-level performance differences. These across-configuration analyses were summarized using medians and interquartile ranges. Statistical significance was defined as two-sided $P < 0.05$.

Model Decision-Making Visualization

To evaluate the networks' learning about glaucoma, we employed Grad-CAM^{++}[44] to visualize the focus areas in fundus images that contribute most to glaucoma prediction. This technique generates class-specific heatmaps, highlighting areas that significantly influence the model's decision-making process. Grad-CAM^{++} was applied to target domain test images from various source–target data pairs using standard DL models, our proposed UDA models, and ideal (train-on-target) models. In addition to qualitative visualizations, we quantitatively assessed the Grad-CAM^{++} results to evaluate model interpretability. For each target test dataset, Grad-CAM^{++} heatmaps were manually reviewed to determine whether the region of highest activation (red area) overlapped with the ONH. Each image with overlapping activation was assigned a value of 1, and nonoverlapping images were assigned a value of 0. For each model, ONH reliance was quantified as the percentage of test images in which the highest Grad-CAM^{++} activation region overlapped with the ONH region.

Implementation Details

Models were implemented in PyTorch. We conducted 20 hyperparameter searches to optimize all the trained baselines and UDA models. The best baseline models were selected based on the maximum validation accuracy over hyperparameters for the glaucoma classification task. The best UDA models were selected based on balancing the maximum source validation accuracy in the glaucoma classification task and minimum target validation accuracy in the domain classification task over hyperparameters. All models were pretrained using ImageNet data and then further fine-tuned on fundus data for the main task. The network architecture was treated as a hyperparameter and selected from ResNet-18 or ResNet-50. We optimized the cross-entropy loss function for all models using the stochastic gradient descent (SGD) algorithm. Images were resized to 224 $\times$ 224 pixels. Other hyperparameters included learning rate ($\{10^{-2}, 10^{-3}, 10^{-4}\}$), batch size ($\{32, 64, 128\}$), weight decay for l_2 regularization ($\{0, 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 5 \times 10^{-4}\}$), and data augmentation techniques ({random horizontal flip, random rotation, random translation}). Rotation angle was randomly selected from {0°, 90°, 180°,

270°}, and images were randomly translated by up to 65 pixels in both x and y directions. Following Ganin and Lempitsky,[37] the GRL coefficient was updated during training according to (Equation 2).

$$\lambda_p = \frac{2}{1 + \exp(-\gamma \cdot p)} - 1 \quad (2)$$

where p denotes normalized training progress from 0 to 1, and γ controls how rapidly the adversarial signal increases during training. We tuned γ over the range $\{2, 3, \ldots, 10\}$.

Results

Glaucoma Classification

Table 3 presents the glaucoma classification performance across full-view fundus source–target data pairs, comparing the proposed UDA method with baseline models. The first row shows the standard DL classification accuracies, the middle row shows the UDA results, and the last row represents the ideal model performance (train-on-target). The standard DL models achieved accuracy rates ranging from 54.3% (95% CI, 52.1%–56.3%) to 71.8% (95% CI, 71.5%–72.1%). The UDA models achieved accuracy rates ranging from 76.0% (95% CI, 73.9%–78.2%) to 90.0% (95% CI, 87.5%–95.0%). The ideal DL models, trained solely on the target domain with class labels provided, achieved accuracy rates ranging from 86.8% (95% CI, 85.7%–88.2%) to 97.7% (95% CI, 97.2%–97.8%).

DeLong's test results for AUC comparisons across full-view fundus source–target pairs are reported in Table 4. The $\Delta_{(UDA-SDL)}$, $\Delta_{(Ideal-UDA)}$, and $\Delta_{(Ideal-SDL)}$ ranged from 0.5 to 35.2, 0.4 to 12.8, and 0.9 to 43.0 percentage points, respectively.

Similarly, Table 5 presents the glaucoma classification performance across various source–target data pairs using ONH-cropped fundus images,

Table 5. Glaucoma Classification Accuracy Across Various ONH-Cropped Fundus Source–Target Data Pairs, Comparing Our Proposed UDA Method With Baseline Models

Method	Source / Target	UIC_{Crop} RIMONE-DL	RIMONE-DL UIC_{Crop}
Standard DL, %		66.2 (62.5–70.3)	70.5 (68.9–73.1)
UDA (proposed), %		82.4 (79.7–85.9)	76.7 (74.8–79.0)
Ideal (train-on-target), %		91.8 (90.6–95.3)	78.3 (76.5–80.7)

Table 3. Glaucoma Classification Accuracy Across Various Full-View Fundus Source–Target Data Pairs, Comparing Our Proposed UDA Method With Baseline Models

Method	Source / Target	UIC LAG	UIC REFUGE	LAG UIC	REFUGE UIC	REFUGE LAG	LAG REFUGE
Standard DL, %		71.8 (71.5–72.1)	70.0 (65.0–77.5)	65.1 (63.0–67.2)	54.3 (52.1–56.3)	62.0 (61.7–62.3)	60.0 (52.5–67.5)
UDA (proposed), %		85.4 (85.2–85.6)	90.0 (87.5–95.0)	81.4 (79.8–83.2)	76.0 (73.9–78.2)	78.0 (77.7–78.3)	90.0 (87.5–95.0)
Ideal (train-on-target), %		97.7 (97.2–97.8)	92.0 (90.0–95.0)	86.8 (85.7–88.2)	86.8 (85.7–88.2)	97.7 (97.2–97.8)	92.0 (90.0–95.0)

Table 4. DeLong Test Results for AUC Comparisons Across Various Full-View Fundus Source–Target Data Pairs, Comparing Our Proposed UDA Method With Baseline Models

Source	Target	AUC (%) SDL	UDA	Ideal	$\Delta_{(UDA-SDL)}$	$\Delta_{(Ideal-UDA)}$	$\Delta_{(Ideal-SDL)}$
UIC	LAG	80.5	92.5	99.6	11.9[a]	7.1[a]	19.0[a]
UIC	REFUGE	87.3	92.5	97.3	5.1[a]	4.8[a]	9.9[a]
LAG	UIC	81.9	84.6	91.3	2.7[a]	6.6[a]	9.3[a]
REFUGE	UIC	48.2	83.5	91.3	35.2[a]	7.8[a]	43.0[a]
REFUGE	LAG	74.6	86.8	99.6	12.2[a]	12.8[a]	25.0[a]
LAG	REFUGE	96.3	96.8	97.2	0.5[a]	0.4[a]	0.9[a]

SDL, standard deep learning.
[a]Denotes DeLong's test $P < 0.0001$.

Table 6. DeLong Test Results for AUC Comparisons Across Various ONH-Cropped Fundus Source–Target Data Pairs, Comparing Our Proposed UDA Method With Baseline Models

		AUC (%)					
Source	Target	SDL	UDA	Ideal	$\Delta_{(UDA-SDL)}$	$\Delta_{(Ideal-UDA)}$	$\Delta_{(Ideal-SDL)}$
UIC_{Crop}	RIMONE-DL	76.6	88.9	97.7	12.3[a]	8.8[a]	21.1[a]
RIMONE-DL	UIC_{Crop}	79.2	83.1	85.3	3.9[a]	2.1[a]	6.0[a]

[a]Denotes DeLong's test $P < 0.0001$.

Table 7. Wilcoxon Signed-Rank Test Results for Paired Performance Differences Across All Source–Target Data Pairs, Comparing UDA, Standard Deep Learning, and Ideal Models

Metric	Comparison	Median	IQR (Q1, Q3)	Wilcoxon Statistic
Accuracy	$\Delta_{(UDA-SDL)}$	0.19[a]	0.09 (Q1 = 0.15, Q3 = 0.24)	0.00
	$\Delta_{(Ideal-UDA)}$	0.12[a]	0.13 (Q1 = 0.06, Q3 = 0.20)	0.00
	$\Delta_{(Ideal-SDL)}$	0.33[a]	0.16 (Q1 = 0.26, Q3 = 0.41)	0.00
AUC	$\Delta_{(UDA-SDL)}$	0.12[a]	0.27 (Q1 = 0.05, Q3 = 0.32)	4.00
	$\Delta_{(Ideal-UDA)}$	0.13[a]	0.13 (Q1 = 0.07, Q3 = 0.20)	0.00
	$\Delta_{(Ideal-SDL)}$	0.36[a]	0.26 (Q1 = 0.21, Q3 = 0.47)	0.00

IQR, interquartile range; Q, quartile.
[a]Denotes Wilcoxon signed-rank test $P < 0.0001$.

comparing the proposed UDA method with baseline models. The standard DL models achieved accuracy rates of 66.2% (95% CI, 62.5%–70.3%) and 70.5% (95% CI, 68.9%–73.1%). The UDA models achieved accuracy rates of 82.4% (95% CI, 79.7%–85.9%) and 76.7% (95% CI, 74.8%–79.0%). The ideal DL models, trained and tested on the target domain with class labels provided, achieved accuracy rates of 91.8% (95% CI, 90.6%–95.3%) and 78.3% (95% CI, 76.5%–80.7%). Additionally, Supplementary Tables S8 to S11 summarize glaucoma classification performance using sensitivity, precision, F1-score, and AUC metrics across the same source–target data pairs, comparing the proposed UDA method with baseline models.

In addition, for the UIC dataset, we validated the standard DL, UDA, and ideal models using expert manual glaucoma grading labels for all source–target pairs, with the corresponding results provided in Supplementary Table S12. Further results for additional source–target dataset combinations are provided in Supplementary Tables S1 and S3. DeLong's test results for AUC comparisons are reported in Table 6 for ONH-cropped fundus pairs. The $\Delta_{(UDA-SDL)}$, $\Delta_{(Ideal-UDA)}$, and $\Delta_{(Ideal-SDL)}$ ranged from 3.9 to 12.3, 2.1 to 8.8, and 6.0 to 21.1 percentage points, respectively.

Further, Table 7 presents Wilcoxon signed-rank results across all source–target pairs, with median paired differences of 0.19, 0.12, and 0.33 for accuracy and 0.12, 0.13, and 0.36 for AUC, respectively. Further, DeLong's test results for AUC comparisons are reported in Supplementary Tables S2 and S4 for various full-view and ONH-cropped fundus pairs.

Generalization to Unseen Domains

Table 8 presents the glaucoma classification performance across various full-view source–target data pairs, comparing the generalizability of our proposed UDA method against baseline models on a third, unseen domain dataset. As summarized in Table 8, the standard DL models achieved accuracy rates of 54.3% and 70.0%, with 95% CIs of 52.1% to 56.3% and 65.0% to 77.5%, respectively. The UDA models demonstrated improved performance, with accuracy rates of 75.4% and 80.6%, with 95% CIs of 75.2% to 75.8% and 79.0% to 83.2%, respectively. The source + target model, baseline trained on the collective source and target data, achieved accuracy rates of 77.5% and 92.0%, with 95% CIs of 75.6% to 79.8% and 90.0% to 95.0%, respectively. DeLong's test results for AUC comparisons in the third, unseen-domain analysis are reported in Table 9. The $\Delta_{(UDA-SDL)}$, $\Delta_{(Source+Target-UDA)}$, and $\Delta_{(Source+Target-SDL)}$ range from 0.8 to 22.6, –2.6 to 11.4, and 0.3 to 34.0 percentage points, respectively.

ONH Reliance in Glaucoma Classification

Table 10 provides a quantitative analysis of Grad-CAM^{++} results across different source–target datasets,

Table 8. Glaucoma Classification Accuracy Performance across Full-View Source–Target Data Pairs, Comparing the Generalizability of Our Proposed UDA Method Against Baseline Models on a Third, Unseen Domain Dataset

Source	Target	Test (Unseen Target)	Standard DL, %	UDA (Proposed), %	Source + Target, %
LAG	REFUGE	UIC	65.1 (63.0–67.2)	80.6 (79.0–83.2)	77.5 (75.6–79.8)
REFUGE	LAG	UIC	54.3 (52.1–56.3)	68.2 (66.4–70.6)	
UIC	LAG	REFUGE	70.0 (65.0–77.5)	80.0 (75.0–85.0)	92.0 (90.0–95.0)
LAG	UIC	REFUGE	60.0 (52.5–67.5)	78.0 (72.5–82.5)	
UIC	REFUGE	LAG	71.8 (71.5–72.1)	75.4 (75.2–75.8)	79.3 (79.1–79.6)
REFUGE	UIC	LAG	62.0 (61.7–62.3)	77.0 (76.7–77.2)	

Table 9. DeLong Test Results for AUC Comparisons Across Full-View Fundus Source–Target Data Pairs, Comparing the Generalizability of the Proposed UDA Method Against Baseline Models on a Third, Unseen-Domain Dataset

Source	Target	Test (Unseen Target)	AUC (%)					
			SDL	UDA	S+T	$\Delta_{(UDA-SDL),\ P}$	$\Delta_{(S+T-UDA)\,P}$	$\Delta_{(S+T-SDL),\,P}$
LAG	REFUGE	UIC	81.9	84.8	82.2	2.8 (0.38)	−2.6 (0.45)	0.3 (0.94)
REFUGE	LAG	UIC	48.2	70.8	82.2	22.6 (<0.001)	11.4 (0.01)	34.0 (<0.001)
UIC	LAG	REFUGE	87.4	94.9	97.6	7.5 (0.13)	2.7 (0.23)	10.2 (0.04)
LAG	UIC	REFUGE	96.3	97.1	97.6	0.8 (0.68)	0.5 (0.77)	1.3 (0.61)
UIC	REFUGE	LAG	80.6	82.2	85.5	1.7 (0.28)	3.2 (0.01)	4.9 (<0.001)
REFUGE	UIC	LAG	74.6	83.1	85.5	8.4 (<0.001)	2.4 (0.05)	10.8 (<0.001)

Δ values represent absolute AUC differences in percentage points. *P* values were calculated using DeLong's test. S+T, source + target model.

Table 10. ONH Reliance Quantified as the Percentage of Test Images in Which the Highest Grad-CAM^{++} Activation Region Overlapped With the ONH Region, Comparing Standard DL, Proposed UDA, and Ideal Models Across Full-View Source–Target Datasets

Method	Source Target	UIC LAG	UIC REFUGE	LAG UIC	REFUGE UIC	REFUGE LAG	LAG REFUGE
Standard DL, %		72.7	83.7	62.5	57.0	67.9	67.3
UDA (proposed), %		45.4	71.4	52.3	75.0	32.1	73.5
Ideal (train-on-target), %		53.7	83.7	91.4	91.4	53.7	83.7

considering both full-view and ONH-cropped images. This table reports the percentage of images, across all test cases, in which each model's prediction was primarily based on features inside the ONH region. Additionally, we summarized the corresponding quantitative results for ONH-cropped test cases, as well as solely for correctly classified predictions across all source–target pairs in Supplementary Tables S13, S14, and S15. For instance, as Table 10's second column shows, on the LAG target dataset, the standard DL model relied on ONH features for 72.7% of images, the UDA model for 45.4%, and the ideal trained-on-target model for 53.7%.

Further, Figure 3 presents Grad-CAM^{++} visualizations for glaucoma classification using a selection of randomly chosen fundus images from the UIC and REFUGE test sets, comparing the heatmap activations of the standard DL model, the proposed UDA model, and the ideal target-trained model. In these visualizations, the red regions of the Grad-CAM^{++} heatmap indicate the primary areas contributing to the model's prediction. Similarly, Grad-CAM^{++} saliency visualizations for LAG, UIC_{Crop}, and RIMONE-DL datasets are provided in Supplementary Figures S1 to S3.

Discussion

We developed a UDA framework to address the well-recognized lack of generalizability in standard

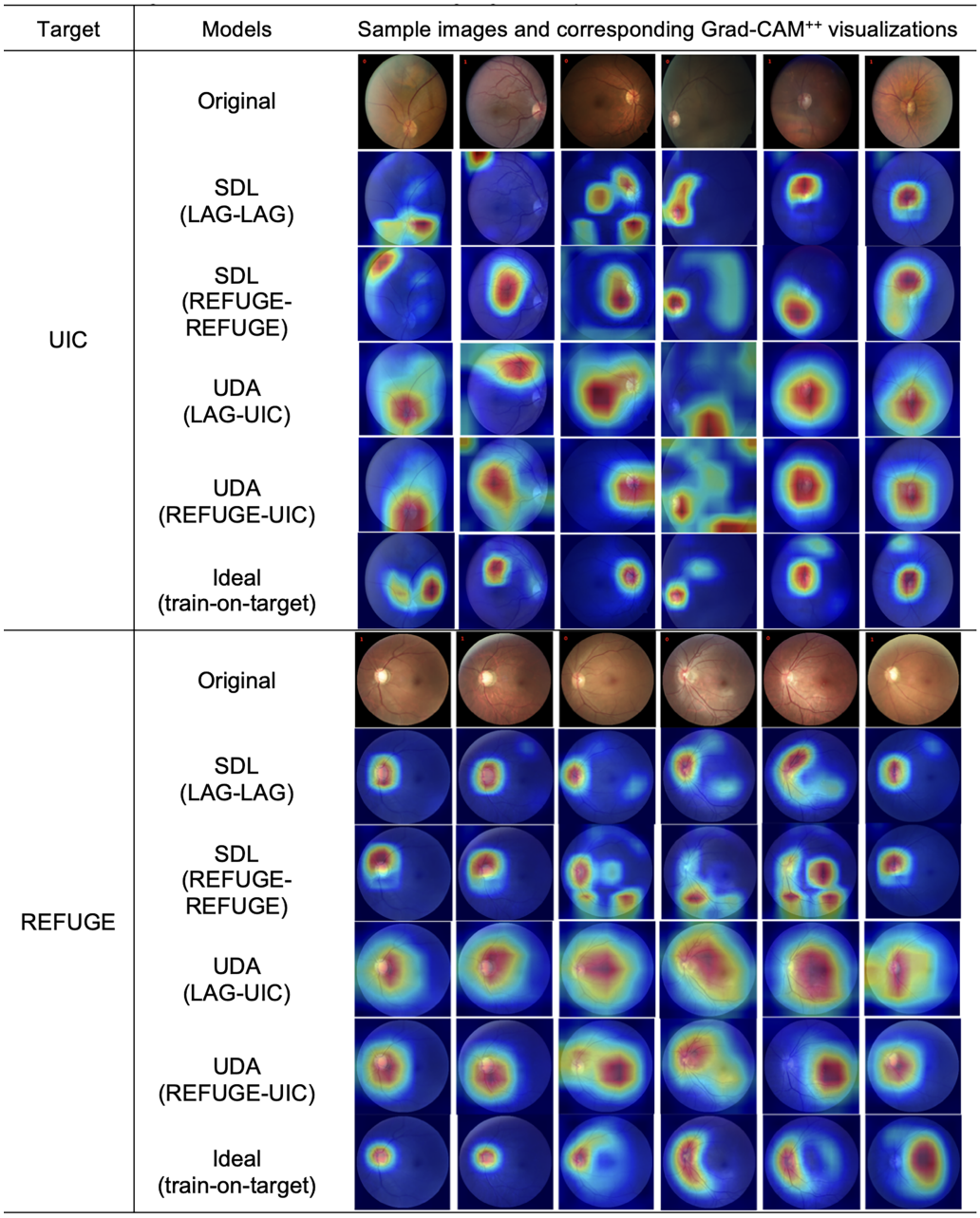


Figure 3. Grad-CAM^{++} visualizations illustrating glaucoma classification predictions generated by the standard DL model, the proposed UDA model, and the ideal model. Representative randomly selected fundus images from the UIC and REFUGE test sets are shown; *red regions* indicate areas most contributing to glaucoma prediction.

DL models for glaucoma classification under domain shift to potentially improve the reliability of DL models for glaucoma detection across heterogeneous imaging sources and clinical environments. Building on adversarial learning principles, our approach mitigates domain-specific biases while maintaining glaucoma classification performance using labeled source and unlabeled target data. Our study yields three main findings. (1) The proposed UDA framework substantially improves the generalizability of glaucoma classification compared with standard DL models commonly used for this task. (2) The UDA method showed consistent paired performance differences across five fundus datasets and 20 distinct source–target domain pairs, supported by nonparametric across-configuration and image-level paired bootstrap analyses. (3) The UDA approach markedly narrows the performance gap to the ideal (train-on-target) models, where ideal baselines serve as an upper performance bound for the task and are not feasible in

real-world settings, as target domain labels are typically unavailable.

Our UDA method enhances the standard DL model's ability to generalize across different datasets, addressing the challenges posed by domain shifts in medical imaging. As shown in Tables 3, 5, and Supplementary Tables S1 and S3, across the evaluated source–target configurations, the UDA method showed consistent paired gains over standard DL, with a median accuracy difference of 0.19 and a median AUC difference of 0.12 by Wilcoxon signed-rank analysis. Across all 20 cross-dataset combinations, accuracy differences ranged from 0.4% to 40.0%, and image-level paired bootstrap analyses provided configuration-specific confidence intervals for these paired differences. For example, the UDA approach showed accuracy differences of 13.6% for UIC-LAG, 20.0% for UIC-REFUGE, 30.0% for LAG-REFUGE, and 17.8% for RIMONE-DL-LAG, illustrating cross-domain generalization across both public and institutional datasets. Per-configuration DeLong analyses further supported the AUC comparisons by evaluating paired model predictions within the same target test sets, avoiding inference based on pooled averages across heterogeneous configurations. Overall, these findings support prior evidence that standard DL models for glaucoma diagnosis can show reduced generalizability under domain shift.[18,25,28,29,42,45] In contrast, our UDA method effectively addresses this issue, reducing the performance gap between standard DL models and the ideal upper-bound models by an average of 71.2%, 49.1%, 87.4%, 38.6%, and 47.0% in accuracy on the UIC, REFUGE, LAG, RIMONE-DL, and UIC_{Crop} target datasets. Variability in image quality and acquisition conditions, particularly within the UIC dataset, is an inherent component of domain shift and reflects the diversity encountered in real-world clinical imaging. Because the purpose of this study is to evaluate domain adaptation performance under realistic and heterogeneous conditions, such variability was intentionally preserved rather than minimized. The consistency of results across datasets suggests that the conclusions are unlikely to be driven by any single dataset or imaging characteristic.

An asymmetric pattern was observed for REFUGE: models tested on REFUGE generally achieved strong performance across multiple source domains, whereas models trained on REFUGE showed more limited generalization to other target datasets. One possible explanation is that REFUGE represents a relatively standardized benchmark domain with more homogeneous imaging characteristics, while datasets such as UIC, LAG, and RIMONE-DL may contain broader variability in image quality, acquisition conditions, disease appearance, and preprocessing. As a result, models trained on more heterogeneous datasets may learn features that transfer well to REFUGE, whereas REFUGE-trained models may not capture the full range of variability present in other datasets. This may also explain the high performance of the UIC + LAG–trained model when REFUGE was used as the unseen test dataset, because combined training on two more heterogeneous domains may provide broader feature coverage than training on REFUGE alone. This asymmetry highlights that strong performance on a widely used benchmark dataset does not necessarily imply robust generalization when that dataset is used as the source domain.

We acknowledge that the UIC dataset used ICD-based billing codes to define glaucoma status, which may introduce misclassification compared with expert clinical grading. To assess the potential impact of this label noise and evaluate the robustness of our findings, we performed sensitivity analyses comparing model performance using ICD-based versus expert manual glaucoma labels for the UIC test set (Supplementary Table S12) and comparing results obtained using cropped ONH regions versus full-fundus images (Supplementary Tables S1, S3, S10, and S11). These analyses showed that our proposed UDA method continued to provide meaningful improvements over standard DL across all source–target combinations involving UIC, indicating that our conclusions remain stable under different label definitions and image preprocessing conditions. Additionally, although labeling criteria and dataset sizes varied across the five datasets, the paired analyses showed the same direction of UDA-versus-standard DL performance differences across source–target configurations, suggesting that the observed results were not driven by a single dataset or source–target pairing. These findings suggest that, although ICD-based labeling may introduce some noise, the framework remains robust when benchmarked against expert-verified outcomes and across heterogeneous datasets with differing label definitions and sample sizes.

Furthermore, we showed that our proposed UDA method not only enhances glaucoma classification accuracy under domain shift but also improves the generalizability of standard DL models when tested on an unseen target dataset. As shown in Table 8, when UIC serves as the unseen target, the UDA method increases generalization accuracy by an average of 14.7%. For REFUGE and LAG as the unseen targets, it improves accuracy by 14% and 9.3%, respectively. Additionally, compared to the upper-performance bound achieved by the collective source + target models, our UDA method closes the performance gap

between standard DL models and the upper-bound models by 82.6%, 52.0%, and 75.0% on the UIC, REFUGE, and LAG target datasets, respectively.

Furthermore, comparing the decision-making process across standard DL, UDA, and ideal models, we found that while the standard DL model heavily relies on ONH region features, the ideal model demonstrates minimal reliance. The UDA model, though not entirely independent, showed reduced ONH reliance compared with the standard DL model, aligning more closely with the ideal model. Quantifying this, as indicated in Table 10 and Supplementary Table S13 for all test images and in Supplementary Tables S14 to S15 for solely correct predictions, we analyzed the overlap between the highest probability region in Grad-CAM^{++} heatmaps and the ONH region. Qualitative interpretability results are reported in Figure 3 and Supplementary Figures S1 to S3, providing cross-dataset visualization of model attention patterns. Given the heterogeneity of the source–target configurations, we interpret the relationship between ONH reliance and model performance descriptively. Although the ONH is clinically central to glaucoma assessment, excessive ONH-focused saliency may indicate reliance on domain-specific ONH appearance, cropping patterns, illumination, image quality, or acquisition-related characteristics rather than domain-stable glaucoma features. In contrast, extra-ONH regions may contain complementary information relevant to glaucoma classification, including peripapillary retinal texture, vessel caliber, vessel trajectory, vascular attenuation, and broader retinal background patterns that may reflect structural or vascular changes associated with glaucomatous damage. By reducing domain-specific feature reliance, the UDA framework may encourage the model to use a broader and more transferable set of retinal features rather than overfitting to ONH-centered dataset characteristics. Across all test images and correctly classified images, the UDA model showed ONH reliance patterns closer to the ideal model than the standard DL model. Averaged across source–target domain pairs, the UDA model's ONH reliance differed from the ideal baseline by 13.78% for correctly classified images and 14.81% for all test images, whereas the corresponding deviations for the standard DL model were 24.28% and 15.82%, respectively. These results suggest that the UDA framework may reduce reliance on domain-specific ONH-associated features while improving cross-domain classification performance.

While our UDA method effectively addresses the performance degradation observed in standard DL models under domain shifts in glaucoma classification, it has limitations. Despite notable improvements, a performance gap remains between our UDA approach and the ideal models, indicating room for further enhancement. While our proposed UDA method reduces this gap by up to 71.2%, a residual gap remains, particularly under large domain shifts. Although ideal model performance may not be fully achievable in real-world settings due to the lack of labeled target data, future work should aim to further narrow this gap by incorporating multisource domain adaptation, self-supervised pretraining, or domain-invariant feature disentanglement to enhance domain invariance and cross-domain generalization. In addition, future studies should explore the use of more advanced backbone architectures, including transformer-based models, ConvNeXt-based models, and retinal foundation models such as RETFound. Although ResNet was used as the feature extraction backbone in this study because it remains a widely adopted and well-benchmarked CNN architecture for glaucoma classification, the proposed UDA framework is not limited to a ResNet backbone. The feature extractor could be replaced by a pretrained retinal foundation model, with the classification head and domain discrimination head retained within the same adversarial adaptation framework. Such integration may further improve domain-invariant feature learning and cross-domain generalization, particularly when large-scale self-supervised or multisource pretrained models are available.

Additionally, while our UDA method demonstrates robustness under significant domain shifts, such as adapting between cropped and full-view fundus images, it does not guarantee complete domain invariance, suggesting that substantial domain differences may still impact performance. Thus, further analysis is needed to explore how to enhance domain invariance across more varied shifts. Moreover, since our UDA approach is trained exclusively on source and target domain data, its generalizability to unseen second-domain data may be limited. To this end, we plan to develop DL models that can ensure generalization across multiple unseen target domains in our future analysis. Future work should also aim to systematically characterize domain shift factors, including the role of potential sources such as patient demographics, imaging devices, acquisition protocols, and differences in imaging color, angle, and pose. Although demographic metadata (e.g., age and sex) were available only for the UIC dataset and not consistently across public datasets, such variability, as well as unmeasured patient-level clinical factors such as ocular and systemic comorbidities, could contribute to domain shifts and affect model generalizability. Addressing these sources of heterogeneity in future

studies through the integration of richer demographic and clinical information could enhance the interpretability and clinical applicability of DL models for glaucoma diagnosis. Although our study leveraged five datasets, we acknowledge the imbalance in sample sizes, particularly the larger number of images in LAG compared to REFUGE and RIMONE-DL, which may introduce bias. However, paired nonparametric and image-level bootstrap analyses showed consistent performance differences across source–target configurations, suggesting that the findings were unlikely to be driven by a single dataset comparison.

Conclusions

DL has shown promise in glaucoma classification using fundus images, but standard DL models often suffer significant performance degradation and limited generalizability under domain shifts, such as differences in data across institutions, posing risks for real-world deployment. To address this critical gap, we propose a novel UDA framework that employs adversarial training to reduce domain-specific bias while enhancing glaucoma classification performance. Our method was evaluated across five fundus imaging datasets, demonstrating up to a 40.0% improvement in accuracy over standard DL models and closing up to 87.4% of the performance gap relative to ideal models across 20 source–target domain pairs. Additionally, our approach improved generalization to previously unseen target domains by up to 14.7% in accuracy, bridging the gap to ideal performance by up to 82.6%. These findings support UDA as a viable strategy for improving the robustness and clinical utility of DL-based glaucoma classification under a real-world domain shift.

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